MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON
TYPE 6012

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, negative-control grid

See figure 1

Mounting position: Any

Weight: 2.5 ounces (70.9 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Conduction Ec1	Conduction Ec2	Nonconduction ec1	Nonconduction ec2	Ehk
Unit:	V	v	v	V	V	v	v	V
Maximum:	6.9	650	1,300	-10	-10	-200	-100	-100, +25
Minimum:	5.7	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	---	---	0	---	---	---

ABSOLUTE RATINGS:

Parameter:	Rg1	ik	Ik	Ig1	Ig2	TA	tk	F
Unit:	Meg	a	A	mA	mA	°C	---	Hz
Maximum:	2.0	5.0	0.5 (see note 1)	50 (see note 1)	50 (see note 1)	+90	---	60
Minimum:	---	---	---	---	---	-75	30	---
TEST CONDITIONS:	---	---	---	---	---	---	30	60

GENERAL:

Qualification - Required

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Peak emission by voltage drop	3	ib = 12 a	0.65	II	etd	---	12	v
3201	Critical grid voltage for conduction (1)	4,5, 7	Epp = 460 Vac; Rgl = 0.1 Meg; Rp = 3,000 ohms	0.65	II	Eccl	-3.5	-7.5	Vdc
3201	Critical grid voltage for conduction (2)	4,5	Epp = 460 Vac; Rgl = 2.0 Meg; Rp = 3,000 ohms	0.65	II	Eccl	---	-8.5	Vdc
(E) 3201	Critical anode voltage for conduction (1)	4	Ecc = 0; Rgl = 0.1 Meg; Rp = 1,000 ohms	0.65	II	Ebb	---	30	Vdc
(E) 3241	Heater current	-		0.65	II	If	2.35	2.85	Aac
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1336	Heater-cathode leakage	-	Ehk = +50, -100 Vdc	---	---	Ihk	---	30	μAdc
3201	Critical grid voltage for conduction (3)	4,5, 6	Ef = 7.0 Vac; Epp = 460 Vac; Rgl = 2.0 Meg; Rp = 3,000 ohms	---	---	Eccl	---	-13.0	Vdc
---	Operation	2,4	Epp = 1,000 Vac; Rgl = 0.1 Meg; Rb/Ib = 0.5 A	---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group B; Epp = 460 Vac; Ik = 0.5 Adc; ik = 5 a; Rgl = 50,000 ohms, resistance load; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
3201	Critical anode voltage for conduction (1)	-		---	---	Ebb	---	60	Vdc
1231	Peak emission by voltage drop	-		---	---	etd	---	20	v
3201	Critical grid voltage for conduction (1)	-		---	---	Eccl	-3.0	-7.5	Vdc

NOTES:

1. Averaged over any 30-second interval.
2. Tube shall operate for a minimum of 30 seconds without signs of continuous conduction as indicated by an oscilloscope.
3. Preheat tube under test for a minimum of 5 minutes, in addition, heat in test socket for a minimum of 5 seconds prior to application of anode voltage. Tie pin numbers 3 and 8 to pin number 5.
4. Connect pin numbers 2 and 8 to pin number 1.

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NOTES: -Continued

5. Shield anode power supply. Use short shielded anode and grid leads. Use noninductive anode and grid resistors. Connect grid resistor directly at test socket.

6. Preheat for a minimum of 15 minutes at the following conditions:

   Ef = 7.0 Vac; Epp = 117 Vac; Egl = 0; Rgl = 2.0 Meg;
   Rb/Ib = 0.5 amp; time interval between preheating and testing time, 5 seconds maximum.

7. This test is to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

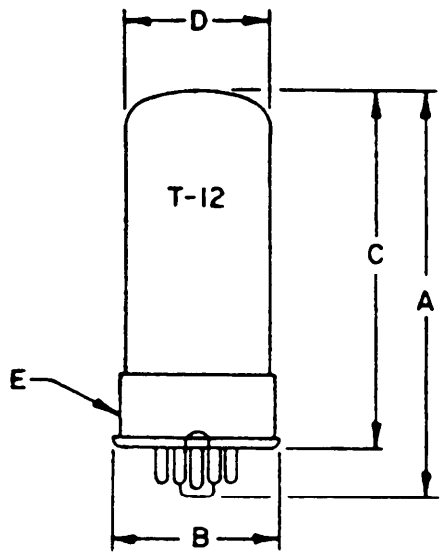
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2932)

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PIN CONNECTIONS	
Pin No.	Element
1	k
2	h
3	g1
5	a
7	h
*8	g2
*Shield grid shall not be used as a control electrode.	

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	---	3.875 (98.43)
B	---	1.720 (43.69)
C	---	3.313 (84.15)
D	1.438 (36.53)	1.562 (39.67)
Ⓔ Quality conformance inspection, part 3 (see note)		
E	Base: B6-100 or B6-161 (EIA)	

NOTE:
Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 6012.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 86 1426

REPLACES EDITION OF 1 OCT 84 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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